

Mathematics

Stage 9

Paper 2

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square brackets in mark scheme

When square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30..... cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

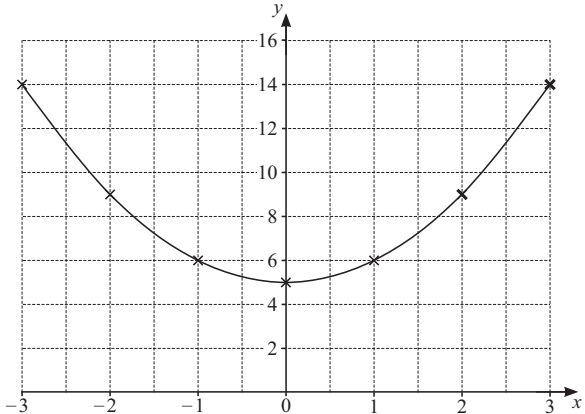
For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

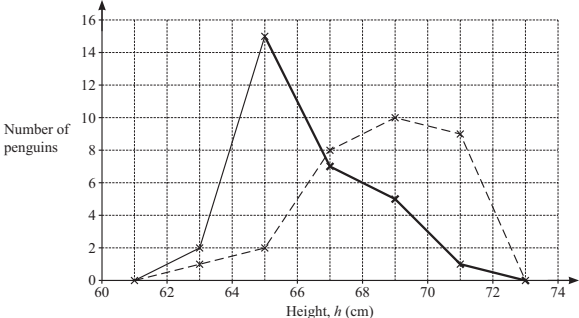
Plotting points

The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance									
1	$(n + 2)^2$ or equivalent	1		Or equivalent, e.g. $n^2 + 4n + 4$									
2	12 (weeks)	1											
3	50	1											
4	An explanation that the highest common factor has not been removed as a factor, e.g. He should also have taken an x out of the bracket.	1		Accept equivalent correct answers, e.g. He should have taken $5x$ out of the bracket. The correct answer is $5x(x - 3)$									
5	(\$) 1.62	2	Award 1 mark for $1.50 \times 1.2 \times 0.9$ or equivalent or for $1.8[0]$	Calculation may be seen in stages for 1 mark.									
6	$(x =)$ 148 $(y =)$ 74	3	Award 2 marks for one correct value. or Award 1 mark for $CAB = 32$ or for ABC or $ACB = 74$										
7	<table><tr><td></td><td>+ve gradient</td><td>–ve gradient</td></tr><tr><td>+ve</td><td>$(y = 2x + 3)$</td><td>$y = 5 - 3x$</td></tr><tr><td>–ve</td><td></td><td>$y = -x - 2$</td></tr></table>		+ve gradient	–ve gradient	+ve	$(y = 2x + 3)$	$y = 5 - 3x$	–ve		$y = -x - 2$	1		Both answers correct for the mark.
	+ve gradient	–ve gradient											
+ve	$(y = 2x + 3)$	$y = 5 - 3x$											
–ve		$y = -x - 2$											

Question	Answer	Marks	Part Marks	Guidance
8	Triangle with vertices at (1, 1), (7, 1) and (1, 9).	2	Award 1 mark for a triangle with two correct vertices or for a correctly enlarged triangle in the wrong position or for correct centre of enlargement but incorrect scale factor >1	
9	Correct curve for $-3 \leq x \leq 3$ drawn 	2	Award 1 mark for 9 and 14 seen or implied or for a correct curve between $(-3, 14)$ and $(1, 6)$.	For 2 marks, ignore work outside of the region $-3 \leq x \leq 3$ For 2 marks, do not accept straight line segments. 1 mark implied by correct points plotted or curve/line through $(2, 9)$ and $(3, 14)$. For this mark, ignore work outside of region $-3 \leq x \leq 1$
10	24250	1		
11	16.4(cm)	2	Award 1 mark for $\frac{36 + 336 + 120}{3 + 21 + 6}$ or better or for a numerator of 492 or for a denominator of 30	Or better, e.g. $\frac{492}{30}$, $492 \div k$

Question	Answer	Marks	Part Marks	Guidance												
12	$-1 < x \leq 3.5$	2	Award 1 mark for $-2 < 2x \leq 7$ or $-3 < x - 2 \leq 1.5$ or either -1 or 3.5 correct with correct inequality signs or both -1 and 3.5 correct but with incorrect inequality signs or equals sign.	Accept $\frac{7}{2}$ or $3\frac{1}{2}$ for 3.5 Accept $\frac{3}{2}$ or $1\frac{1}{2}$ for 1.5												
13(a)	Plots at $(67, 7)$, $(69, 5)$ and $(71, 1)$ and $(73, 0)$ and four solid straight-line segments. 	2	Award 1 mark for one or more of the four plots correct.	Mark intention.												
13(b)	<table border="1"><thead><tr><th>True</th><th>False</th><th>Not enough info</th></tr></thead><tbody><tr><td>✓</td><td></td><td></td></tr><tr><td></td><td></td><td>✓</td></tr><tr><td>✓</td><td></td><td></td></tr></tbody></table>	True	False	Not enough info	✓					✓	✓			2	Award 1 mark for two correct ticks.	Accept any clear indication.
True	False	Not enough info														
✓																
		✓														
✓																

Question	Answer	Marks	Part Marks	Guidance
14	$(x=) 3$	2	Award 1 mark for $10 = 5(x - 1)$ or better.	Or better, e.g. $10 = 5x - 5$ or $x - 1 = 2$
15	17	1		
16	$x^2 - 2x - 8$	2	Award 1 mark for three correct terms from $x^2 + 2x - 4x - 8$	$-2x$ counts as two correct terms.
17(a)	4, 1	2	Award 1 mark for [fourth term =] 4 or for <i>(their</i> fourth term $- 3)^2$ correctly evaluated.	Both answers in the correct order for the mark.
17(b)	multiply by 2 or equivalent	1		Or equivalent, e.g. times 2, $\times 2$, double
18	$(x=) 9$	3	Award 2 marks for $2 \times 0.5 \times 2 \times 1.5 + x(2 + 2.5 + 1.5) = 57$ or equivalent or better or Award 1 mark for $0.5 \times 2 \times 1.5$ and one other area.	Or better, e.g. $\frac{57 - 2 \times 0.5 \times 2 \times 1.5}{2 + 2.5 + 1.5}$ For 2 marks, correct method may be seen in stages. One other area from $0.5 \times 2 \times 1.5$ or $2 \times x$ or $2.5 \times x$ or $1.5 \times x$ 1 mark implied by 54
19	8	1		

Question	Answer	Marks	Part Marks	Guidance
20	3.92 (cm)	3	Award 2 marks for answer in range 3.921 to 3.924 or for $\sqrt{\frac{48.34}{\pi}}$ or Award 1 mark for $\frac{48.34}{\pi}$	Accept values of π between 3.14 and $\frac{22}{7}$
21	34 51 102	3	Award 1 mark for each correct answer. If 0 marks scored, award 1 mark for $12 \div k$ where $k = 9 - 5$ or $9 - 3$ or $5 - 3$ or 4 or 6 or 2	Answers can be given in any order.
22	(y =) 7	1		

Question	Answer	Marks	Part Marks	Guidance
23(a)	3 added to top row of stem-and-leaf diagram (to represent 43)	2	Award 1 mark for 43 seen but not added correctly to the diagram or for getting range correct by adding 9 to bottom row (to represent 89) or for getting median correct by adding any value less than or equal to 64 (other than 43).	
23(b)	(Eva may be correct because) the median for class Z is closer to one minute/60 seconds and (Eva may not be correct because) the range for class Z is bigger suggesting the results are more varied/some in class Z were a long way from one minute.	2	Award 1 mark for each correct answer.	Accept equivalent correct reasons.
24	41.7 to 41.71...(cm)	4	Award 3 marks for $0.5 \times 2 \times \pi \times 5$ or better or Award 2 marks for $AB = 10$ or $\frac{1}{2}AB = 5$ or Award 1 mark for $x^2 + 12^2 = 13^2$ or equivalent or better.	Accept values of π between 3.14 and $\frac{22}{7}$ 3 marks implied by 15.7 to 15.71... Or better, e.g. $\sqrt{13^2 - 12^2}$, $\sqrt{25}$